

A strong dose-response relation between serum concentrations of persistent organic pollutants and diabetes: results from the National Health and Ex...



OBJECTIVE: Low-level exposure to some persistent organic pollutants (POPs) has recently become a focus because of their possible link with the risk of diabetes. RESEARCH DESIGN AND METHODS: Cross-sectional associations of the serum concentrations of POPs with diabetes prevalence were investigated in 2,016 adult participants in the National Health and Nutrition Examination Survey 1999-2002. Six POPs (2,2',4,4',5,5'-hexachlorobiphenyl, 1,2,3,4,6,7,8-heptachlorodibenzo-p-dioxin, 1,2,3,4,6,7,8,9-octachlorodibenzo-p-dioxin, oxychlordane, p,p'-dichlorodiphenyltrichloroethane, and trans-nonachlor) were selected, because they were detectable in $\geq 80\%$ of participants. RESULTS: Compared with subjects with serum concentrations below the limit of detection, after adjustment for age, sex, race and ethnicity, poverty income ratio, BMI, and waist circumference, diabetes prevalence was strongly positively associated with lipid-adjusted serum concentrations of all six POPs. When the participants were classified according to the sum of category numbers of the six POPs, adjusted odds ratios were 1.0, 14.0, 14.7, 38.3, and 37.7 (P for trend < 0.001). The association was consistent in stratified analyses and stronger in younger participants, Mexican Americans, and obese individuals. CONCLUSIONS: There were striking dose-response relations between serum concentrations of six selected POPs and the prevalence of diabetes. The strong graded association could offer a compelling challenge to future epidemiologic and toxicological research.